

Junaid Mansur Ifti

🌐 <https://junaidmifti.github.io>

✉ Email: bsse1027@iit.du.ac.bd

🐙 Github: jmifti

in jmifti

EDUCATION

University Of Dhaka, Bangladesh

2018 - 2022

Bachelor of Science in Software Engineering

CGPA: 3.58/4.00 (3.98/4.00 in Last 2 Years)

Notable Courses: Machine Learning, Artificial Intelligence, Software Requirement Engineering, Software Metrics, Software Security, Software Design and Analysis, Distributed Systems

RESEARCH EXPERIENCE & PUBLICATIONS

Publications under Review

- **Ahmed Ryan, Junaid Mansur Ifti, Akond Ashfaqur Ur Rahman, Md Erfan, Md Rayhanur Rahman.** “Unveiling Malicious Logic: Towards a Statement-Level Taxonomy and Dataset for Securing Python Packages.” *Under Review at IEEE Transactions on Software Engineering (TSE)* (6 Citations).
 - Derived a fine-grained taxonomy of 47 malicious indicators across 370 malicious Python packages to enable behavior-centric detection and train semantic-aware models.
 - Applied sequential pattern mining to uncover recurring indicator sequences that characterize common attack workflows to strengthen software supply-chain defenses.

Independent Researcher

June 2024 – Present

Automated Secret Migration and Benchmarking by LLM | Supervised by Dr. Md Rayhanur Rahman (The University of Alabama)

- Developing an evaluation framework to assess Large Language Models (e.g., GPT-4o, Claude 3.5 Sonnet) on their ability to autonomously remediate hard-coded secrets in vulnerable codebases.
- Constructing a multi-language Golden Solution Corpus from fuzzed vulnerable repositories to measure structural correctness via AST similarity (CodeBERT, Zhang-Shasha) and security remediation via Gitleaks scans.

Undergraduate Researcher, *Distributed Systems & Software Engineering Research Group*, Bangladesh

2022

- Developed “GANomaly”, an ML-based web application to identify anomalous web traffic in Google Analytics data, supervised by Dr. Kazi Muheymin.
- Implemented statistical outlier detection techniques and rule-based algorithms based on the IEEE paper “Using Google Analytics to Support Cybersecurity Forensics” to support cybersecurity forensics.

PROFESSIONAL EXPERIENCE

Ding, Ireland (Hybrid-Remote)

Senior Software Engineer

August 2026 – Present

- Serve as technical lead and team co-supervisor, mentoring junior engineers and accelerating engineering output by integrating AI-assisted coding tools (e.g., Claude Code) into the development workflow.
- Lead critical third-party API integrations, acting as the primary technical liaison between external suppliers and internal C-level directors.
- Spearhead cross-functional initiatives with broader engineering teams to modernize legacy .NET applications, remediate security vulnerabilities, and maintain CI/CD pipeline stability.

Junior Software Engineer

May 2023 – August 2026

- Maintained the backend of internal web applications, developed RESTful APIs, and integrated 3rd-party payment APIs (REST/SOAP) into the core web system.
- Led technical meetings and facilitated knowledge transfer sessions for both technical and non-technical stakeholders in a multicultural environment.

SWE Intern, *Samsung Research & Development Institute, Bangladesh (Onsite)*

January 2022 – June 2022

- Contributed to the core Android codebase (100K+ LOC) using Kotlin and Java, improving and increasing unit test case coverage by 40%.

ACADEMIC & RESEARCH PROJECTS

- **GQM Research: Impact of Physical Health on Developer Productivity** | *Project Link* 2022
Investigated correlations using the Goal Question Metric approach via manually collected data from software engineers.
- **A Tool for Commit Classification** | *Project Link* 2022
Developed a Python tool leveraging source code changes and commit messages to categorize maintenance activities into Corrective, Perfective, and Adaptive classes.
- **Air Quality Index Predictor (LR)** | *Project Link* 2022
Pre-processed scraped custom datasets and applied linear regression to model the correlation between weather patterns and AQI in Chittagong.
- **Control Flow Graph and Software Metrics Tool** | *Project Link* 2022
Built a Python tool to compute Halstead complexity and LOC for automated software quality assessment.
- **Stock Price Prediction (LR)** | *Project Link* 2022
Developed a linear regression model to predict stock market trends using historical data within the context of Bangladesh.
- **AI Game Agents (Connect 4 & Wumpus World)** 2021
Designed inferential logic agents and applied the Minimax algorithm with Alpha-Beta pruning for heuristic evaluation and decision-making.
- **Machine Learning Classifiers From Scratch** 2021
Built a K-Nearest Neighbors (KNN) algorithm to predict divorce likelihood based on psychological datasets, and a Naive Bayes classifier for lightweight skin pixel detection in images.

SKILLS

- **Programming:** Python, C#, C++, Java, JavaScript, PHP, Kotlin
- **Frameworks & Technologies:** .NET Core, Flask, pandas, scikit-learn, matplotlib, MongoDB, MSSQL, NUnit, JUnit
- **Tools & Platforms:** Jenkins, Git, Docker, Docker-Compose, Jupyter Notebook, Nginx
- **Problem Solving Profile:** (Leetcode) bsse1027 | **Solved Problems: 111** | **Max Contest Rating: 1463**

STANDARDIZED TEST SCORES

IELTS Academic: Listening 8.5 | Speaking 8 | Writing 7.5 | Reading 7.5 | **Overall 8**

ACCOMPLISHMENTS & INVOLVEMENT

- **National Cyber Drill by BGD e-Gov CIRT:** Secured a spot in the Top 20 among 234 teams in a national CTF competition (2020).
- **Awards:** Government Scholarship for Higher Secondary (2017) and Secondary Certificate Exam (2015); Bangladesh Physics Olympiad Divisional Medalist (2012).
- **IIT Software Engineers' Community (2019-2022):** Served as Organizing & Cultural Secretary; led technical teams for "ITVerse 1.0" and operational teams for "TechCrunch 2.0," an ADB-funded workshop promoting women in IT.
- **Fight For Life Initiative (2020):** Volunteered to create an integrated payment website to raise emergency funds for poverty-stricken individuals during the initial Covid-19 lockdown.

REFERENCES

Dr. Md Rayhanur Rahman
Assistant Professor, Dept of Computer Science
The University of Alabama, USA

Dr. Kazi Muheymin-Us-Sakib, Professor
Institute of Information Technology
University of Dhaka, Bangladesh
Email: sakib@iit.du.ac.bd

Dr. B M Mainul Hossain, Professor & Director
Institute of Information Technology
University of Dhaka, Bangladesh
Email: mainul@iit.du.ac.bd